

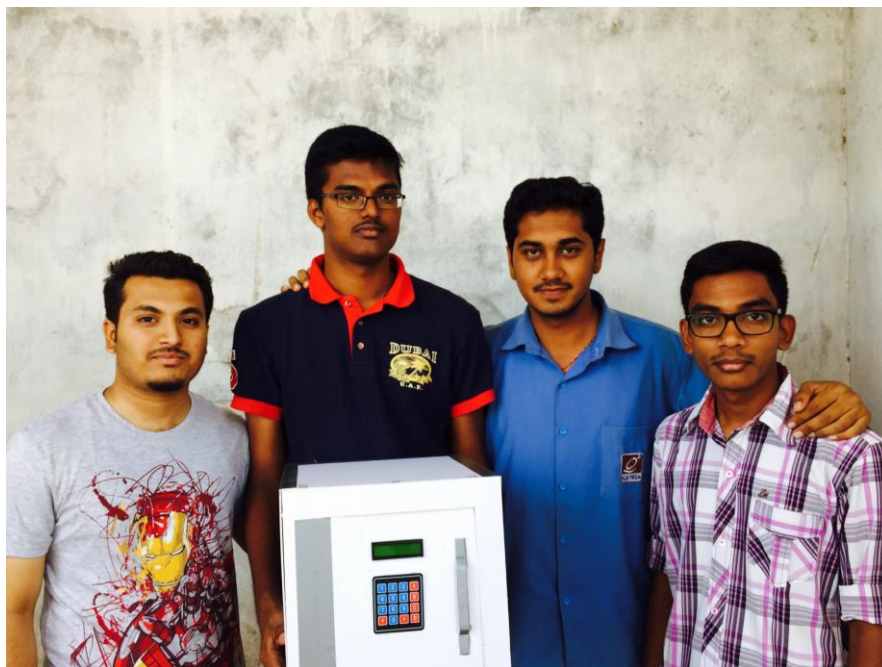


Mechatronics and Control Engineering Course Project Report

Title: Digital Locker

Team members:

NAME	ROLL NO	USN	SIGN
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Problem definition:

Building a digital locker that has a user based interface like password protected locking system, using keypad, arduino and has emergency SMS alert facility in case of theft or forced opening of locker.

Locker should be such that it should have a password protected lock that can be opened using keypad, Bluetooth connected phone or by any other electronic means.

Objectives:

- Building a digital locker by interfacing arduino with a electronic lock
- Interfacing arduino to connectivity devices like Bluetooth
- Interfacing arduino to keypad for password entry
- Interfacing arduino to GSM module for sending emergency SMS alert
- Interfacing arduino to LCD screen for better user experience

Literature Survey:

The oldest known mechanical functioning lock was an Egyptian door lock used about 2000BC, made of wood and fastened vertically on the door post, the wooden block contained moveable pins or “pin tumblers” that dropped by gravity into openings in the cross piece of “bolts” and locked the door. It was operated by a wooden key with pegs that raised the number of tumblers sufficiently to clear the bolt so that it could be pulled back. The major disadvantage with it is that it was wholly made of wood.

With the fast growing of technology, the lock has evolved from only pure mechanical type into electrical and electronic type. With the assemblage of electronic and mechanic at system, the lock has developed into a high-tech style which uses some special authentication method. Usually, it is called electronic lock or smart lock as it can operate based on the given information. In the current market, there are many authentication methods to trigger the electronic lock including, button sequence, radio frequency identification (RFID), biometric, security token and etc. Generally, these kinds of authentication use the embedded system to process or verify password or data entered by the users. The intruders have to spend quite a time to crack these kinds of high tech lock.

The electronic code lock, which is designed to respond to an electronic logic signal mechanism, with a digit sequence counter performing the function of a key. They are operated by inputting the correct code by external means through a keypad into a micro controller which already have a pin number in its EEPROM memory (internal) to compare with, so that if it is correct (that is the pin number inputted by the user), It activates the port to which the lock is connected.

AUTHENTICATION METHODS:

Password Type: Among these authentication methods, the most popular type is using the numerical code as access tool. This method requires the user to enter the correct sequence of code via keypad (button) in order to unlock the system. Besides numerical code, the password

is also one of the methods under the button sequence type. It may consist of the combination of alphabetic and numerical code.

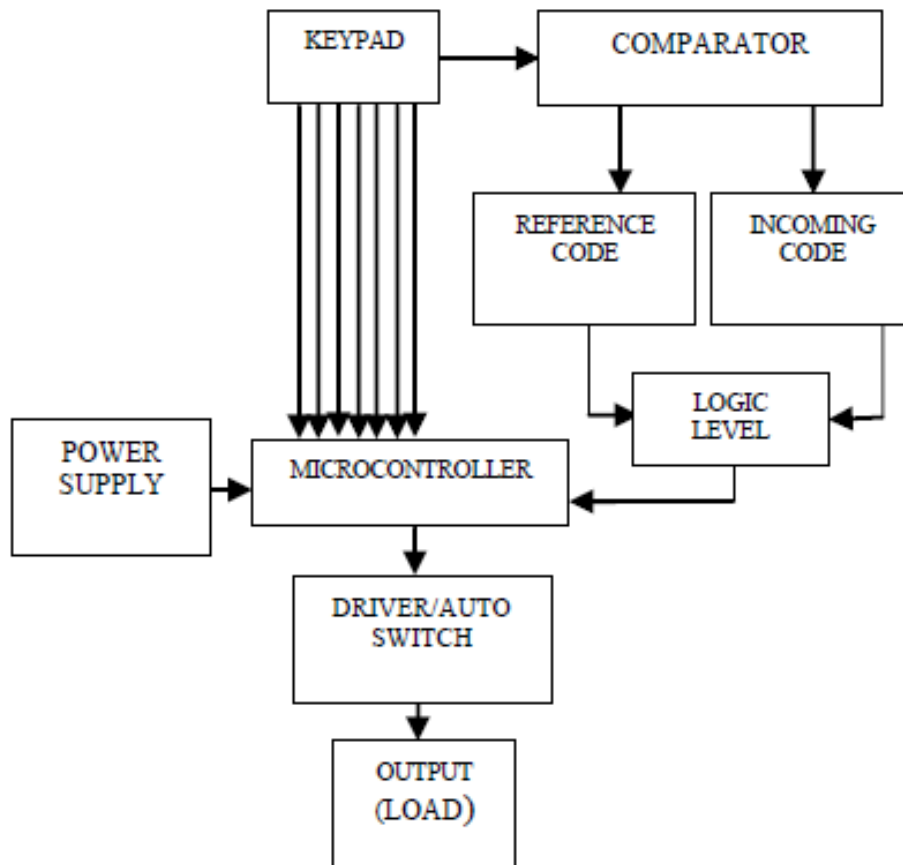
Token Type: For the authentication using security token, it requires possession of the so called 'key' or token in order to unlock. The token can be a card, ring, passport or other devices that contained the information to be verified by the security system. The system needs the tag and the reader to operate. RFID is a method that authenticates the user by using the radio wave which is belongs to the token type group. It needs a tag to store the information as the code and the tag is usually embedded into an object, or living thing including human for the verification.

Biometric Type: the biometrics authentication is the access operation that is growing fast and is getting more popular as it performs in high security level. There are several types of biometric authentication such as fingerprint scanning, iris and retina scanning, voice detection, hand geometry recognition, face detection and so on. This kind of system is said to be safe as each individual having a unique pattern for the retina, iris, and fingerprint as well as other body parts that are used as the authentication method.

The result which we expect from our project is a system that is used to lock and unlock the locker. Rather than using a key, it uses a command in the form of a password .The password would be pre-decided. Due to use of password only authorized users would be allowed to access the locker thus increasing the security level. The use of electronic lock using Bluetooth module and Android smart phones in addition to providing ease of use, also provide better security than conventional key. The system is basically designed to simulate an electronic key, which is controlled through a Wi-Fi-enabled smart phone. Controlling conducted by sending a command via Android smart phone to Wi-Fi module and then to the Microcontroller circuit that acts as a connection between Android smart phone and lock.

Block Diagram:

The following figure shows the basic skeleton of the digital/electronic locking system.



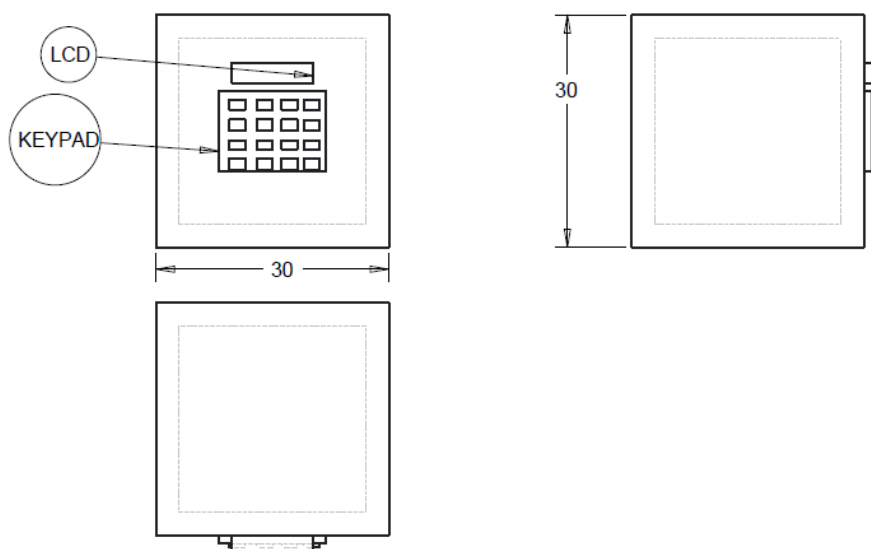
Input to the microcontroller is given through the keypad.

For extra added security blue tooth module (HC-05) can be interfaced with the arduino, which can be connected to the smart phone. In that case the user would have to enter the password/entry key twice, once through keypad and then through their smart phone.

CAD Drawings:**Description of Detailed Design:**

Material used for locker: 3mm Foam board

Lock used: solenoid lock



Bill of materials:

SI NO	COMPONENTS	QUANTITY	COST in Rupees
1.	Arduino Mega	1	800
2.	16*2 LCD	1	140
3.	Keypad 4*4	1	90
4.	Relay Board	1	200
5.	HC-06 Bluetooth module	1	250
6.	GSM 300 Module	1	900
7.	Miscellaneous(wires etc)		100
8.	Foam Board	1	100
9.	Solenoid lock	1	850

Specifications of Components:**1. HC-06 Bluetooth module**

- 1、Bluetooth protocol: Bluetooth Specification v2.0+EDR
- 2、Frequency: 2.4GHz ISM band
- 3、Modulation: GFSK (Gaussian Frequency Shift Keying)
- 4、Emission power: $\leq 4\text{dBm}$, Class 2
- 5、Sensitivity: $\leq -84\text{dBm}$ at 0.1% BER
- 6、Speed: Asynchronous: 2.1Mbps (Max) / 160 kbps, Synchronous: 1Mbps/1Mbps
- 7、Security: Authentication and encryption
- 8、Profiles: Bluetooth serial port
- 9、Power supply: +3.3VDC 50mA
- 10、Working temperature: $-20 \sim +75$ Centigrade
- 11、Dimension: 26.9mm x 13mm x 2.2 mm

2. LCD 16 X 2

1. Built-in controller (ST7066 or Equivalent)
2. +5V power supply only
3. Negative voltage optional for +3V power supply
4. 1/16 duty cycle

3. 4*4 keypad

1. Connector: Dupont 8 pins, 0.1 inch (2.54mm) pitch
2. Peel off the white sticker on the back of the keyboard, can be securely affixed to the surface of the chassis
3. Operational Temperature: -20 to 40 degree Celsius

4. Relay Board

1. 5V 8-Channel interface board and each one needs 15-20mA driver current
2. Equipped with high current relay AC250 10A

3. DC30V 10A
4. Standard interference that can be directly controlled by microcontroller.
5. Indication LED's for relay output status.

5. GSM Module

1. Tri-Band GSM/GPRS 900/1800/1900MHz or
2. Quad-Band GSM/GPRS 850/900/1800/1900MHz
3. GPRS multi-slot class 10/8
4. GPRS mobile station class B
 - Compliant to GSM phase 2/2+
 - Class 4 (2W @ 850/900 MHz)
 - Class 1 (1W @ 1800/1900MHz)
 - Dimension: 33mm x 33mm x 3mm
 - Weight: 7.8 g
 - Control via AT commands (GSM 07.07, 07.05 and SIMCom enhanced AT Commands)
 - SIM application toolkit
 - Supply voltage range 3.4V...4.5V
 - Low power consumption
 - Normal operation temperature: -30 °C to +80 °C
 - Restricted operation temperature: -30 °C to -40 °C and +80 °C to +85 °C
 - Storage temperature: -45°C to +90°C

Specifications for fax

- Group 3, class 1

Specifications for data:

- PBCCCH support
- GPRS class 10: max. 85.6 Kbps (downlink)
- PPP-stack
- Coding schemes CS 1, 2, 3, 4
- CSD up to 14.4 kbps
- USSD
- Non transparent mode

Specifications for SMS via GSM

- Point-to-point MO and MT
- SMS cell broadcast
- Text and PDU mode

Specifications for voice

- Tricodec
 - Half rate (HR)
 - Full rate (FR)
 - Enhanced Full rate (EFR)
- Hands-free operation (Echo suppression)

Interfaces

- Interface to external SIM 3V/1.8V
- Two analog audio interfaces
- RTC backup
- Serial interface and debug interface
- LCD interface
- Keypad interface
- Antenna connector and antenna pad
- GPIO

6. Arduino Mega

Microcontroller	ATmega2560
Operating Voltage	5V
Input Voltage (recommended)	7-12V
Input Voltage (limit)	6-20V
Digital I/O Pins	54 (of which 15 provide PWM output)
Analog Input Pins	16
DC Current per I/O Pin	20 mA
DC Current for 3.3V Pin	50 mA
Flash Memory	256 KB of which 8 KB used by bootloader
SRAM	8 KB
EEPROM	4 KB
Clock Speed	16 MHz
LED_BUILTIN	13
Length	101.52 mm
Width	53.3 mm
Weight	37 g

Input and output functionality testing:

1. Bluetooth

```
String voice;
void setup() {
  pinMode(13, OUTPUT);
  Serial.begin(9600);
}
void loop() {
  while (Serial.available())
  {
    delay(10);
    char c = Serial.read();
    voice += c;
  }
  if (voice.length() > 0)
  {
    Serial.println(voice);
    if(voice == "on")
    {
      digitalWrite(13, HIGH);
    }
    if(voice == "off")
    {
      digitalWrite(13, LOW);
    }
  }
  voice="";
}
}
```

2. LCD display

```
#include <LiquidCrystal.h>

// initialize the library with the numbers of the interface pins
LiquidCrystal lcd(12, 11, 38,39,40,41);

void setup() {
  // set up the LCD's number of columns and rows:
  lcd.begin(16, 2);
  // Print a message to the LCD.
  lcd.print("hello, world!");
}

void loop() {
  // set the cursor to column 0, line 1
  // (note: line 1 is the second row, since counting begins with 0):
  lcd.setCursor(0, 1);
  // print the number of seconds since reset:
  lcd.print(millis() / 1000);}
}
```

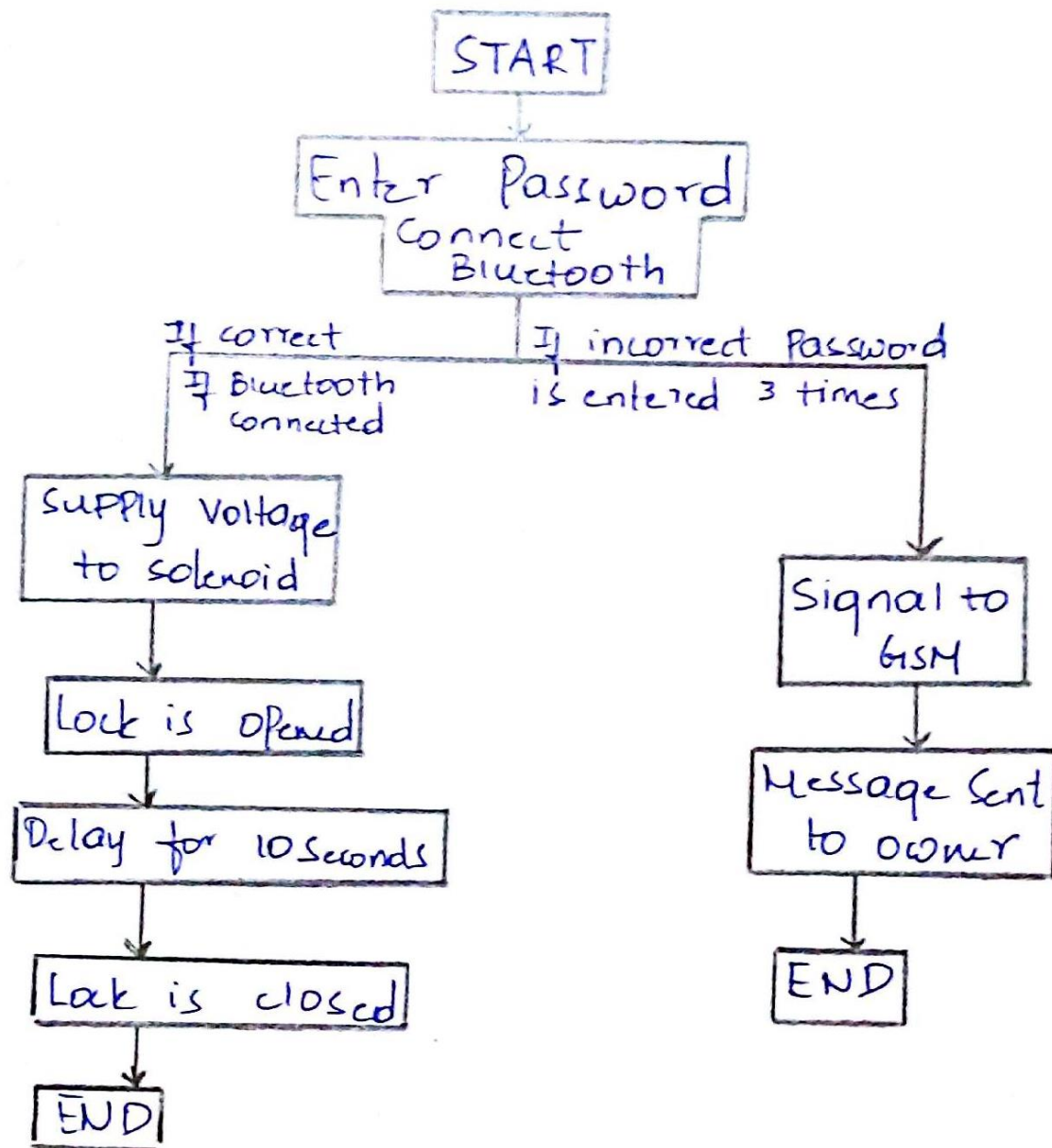
3. Keypad

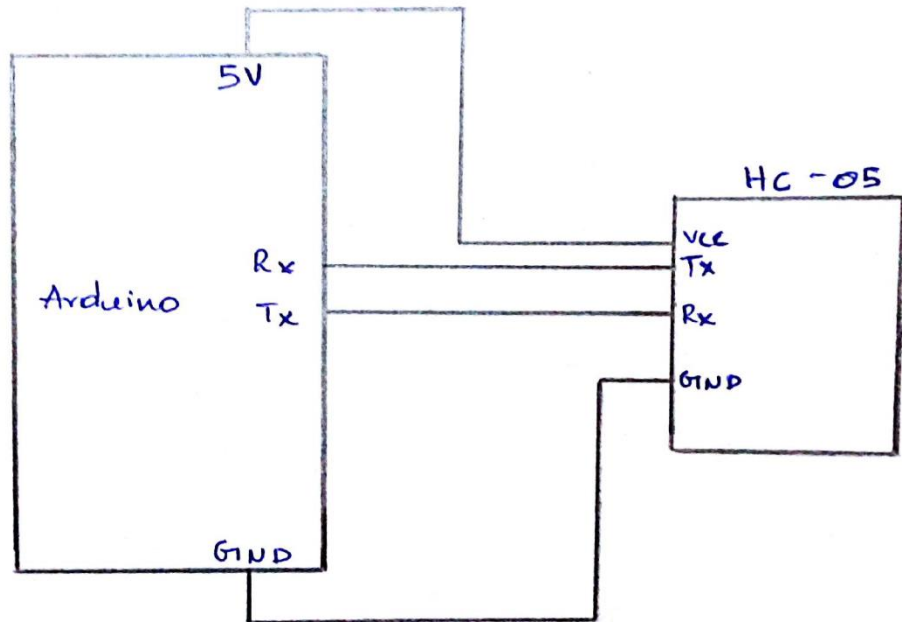
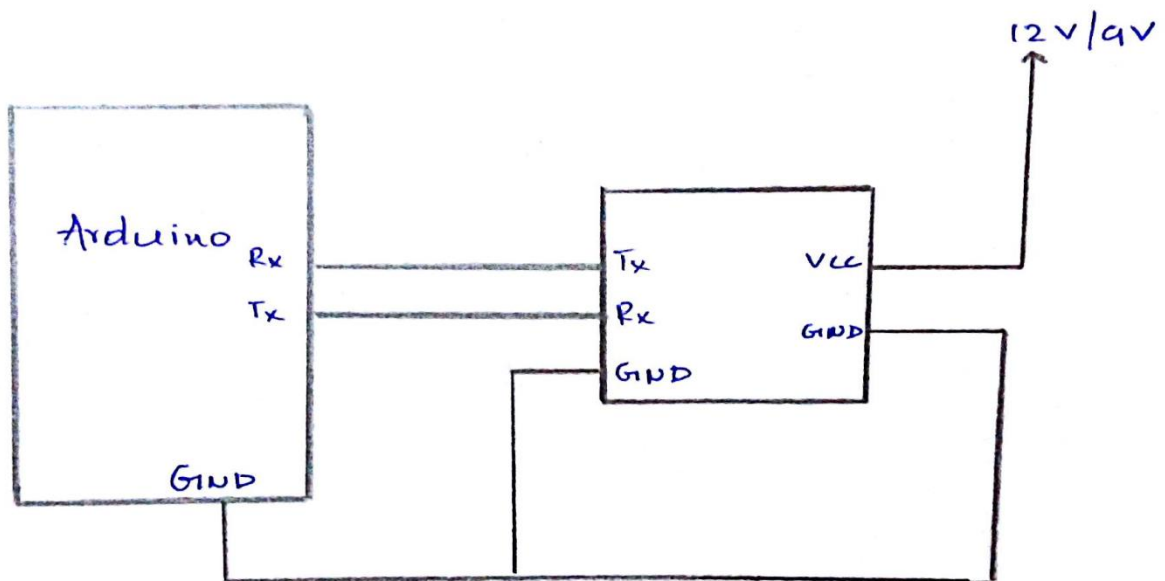
```
//Define the keymap
char keys[ROWS][COLS] = {
  {'1','2','3','A'},
  {'4','5','6','B'},
  {'7','8','9','C'},
  {'*','0','#','D'}
};

//// Connect keypad ROW0, ROW1, ROW2 and ROW3 to these Arduino pins.
byte rowPins[ROWS] = {6,7,8,9}; //connect to row pinouts

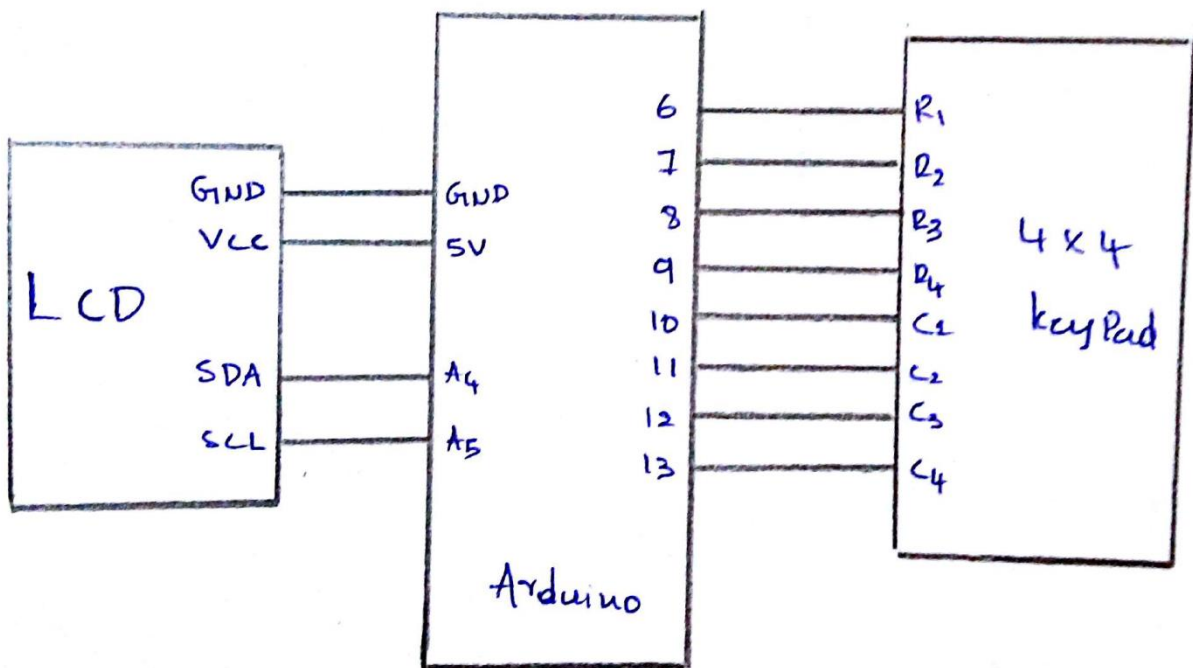
// Connect keypad COL0, COL1, COL2 and COL3 to these Arduino pins.
byte colPins[COLS] = {2,3,4,5}; //connect to column pinouts

// Create the Keypad
Keypad keypad = Keypad( makeKeymap(keys), rowPins, colPins, ROWS, COLS );
LiquidCrystal lcd(12, 11, 38,39,40,41);
void setup(){
  lcd.begin(16,2);
  lcd.setCursor(0,0);
  //lcd.print("digital locker");
  lcd.setCursor(0,0);
  lcd.print("Enter Password");
  //delay(5000);
  //lcd.clear();
  Serial.begin(9600);
  pinMode (IN1, OUTPUT);
  pinMode (IN2, OUTPUT);
}
```

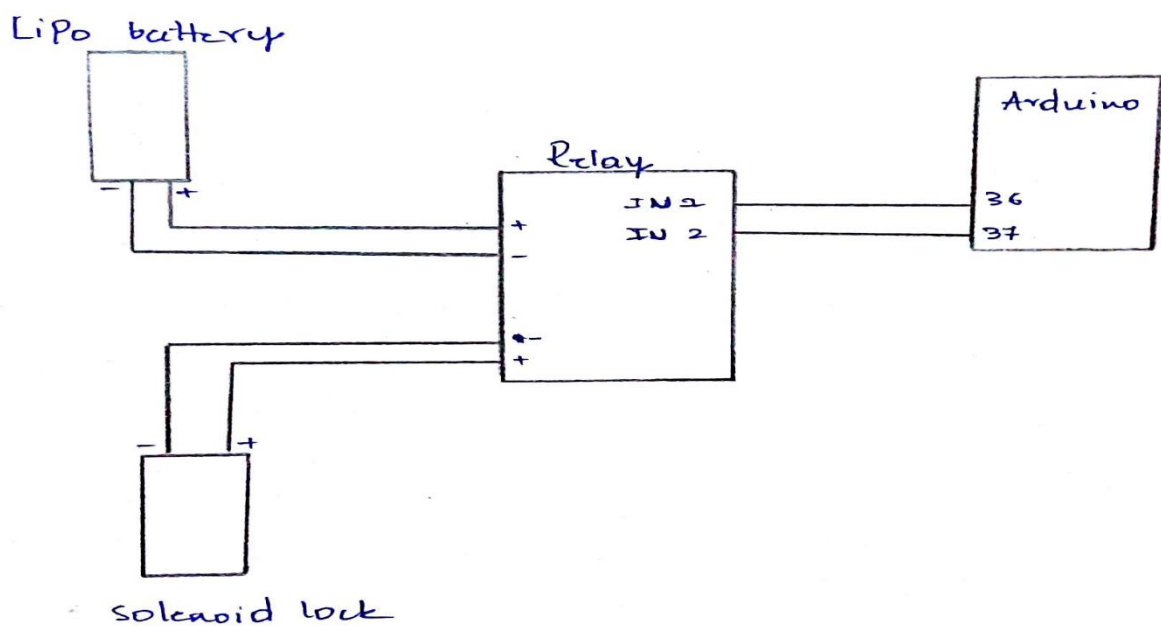
Program Flow Diagram:**GENERAL FLOW DIAGRAM**

Circuit Diagrams:**BLUETOOTH & ARDUINO****GSM MODULE INTERFACE**

LCD KEYPAD INTERFACE



RELAY SOLENOID INTERFACING



Results & Conclusions:

With the help of a microcontroller (Arduino AT mega) we have made a digital locker that has a keypad, Bluetooth, LCD display and GSM module.

The following objectives were completed successfully:

- Building a digital locker by interfacing arduino with a electronic lock
- Interfacing arduino to connectivity devices like Bluetooth
- Interfacing arduino to keypad for password entry
- Interfacing arduino to GSM module for sending emergency SMS alert
- Interfacing arduino to LCD screen for better user experience

Learning experience:

NAME: BINDU SAINATH REDDY K

USN: 01FE15BME039

- Using Arduino AT mega to interface LCD, keypad, Bluetooth
- Using relay to control solenoid
- Installing different libraries to Arduino IDE
- Use of breadboard for circuit connections

Learning experience:

NAME: GURURAJ SHETTY

USN: 01FE15BME058

- Interfacing of Arduino with basic electronic components.
- Use of AT commands in configuring devices
- Understanding of arduino program and modifying it to our needs.
- Use of libraries in arduino software.
- Understanding of battery ratings.

Learning experience:

NAME: KARTIK KULKARNI

USN: 01FE15BME070

- Interfacing Arduino with different components
- Using breadboard to make circuit connections
- Making box/locker
- Fabrication of prototype.

Learning experience:

NAME: KETAN PANDURANGI

USN: 01FE15BME074

- Interfacing of Arduino with basic electronic components.
- Designing of prototype.
- Identifying materials for prototype.
- Team work.
- Time management.
- Fabrication techniques of prototype.
- Programming for specific purposes.

References:

<https://www.arduino.cc/en/main/software>

<http://playground.arduino.cc/Code/Keypad>

<https://www.arduino.cc/en/Reference/LiquidCrystal>

<http://playground.arduino.cc/Code/Password>

<http://www.instructables.com/>

Appendix:

Program:

```
#include <Password.h>
```

```
#include <Password.h>
```

```
#include <LiquidCrystal.h>
```

```
#include <Keypad.h>
```

```
String newPasswordString; //hold the new password
```

```
String voice;
```

```
char newPassword[6]; //character string of newPasswordString
```

```
int IN1 = 36;
```

```
int IN2 = 37;
```

```
//initialize password to xxxx
```

```
//you can use password.set(newPassword) to overwrite it
```

```
Password password = Password( "0000" );
```

```
byte maxPasswordLength = 4;
```

```
byte currentPasswordLength = 0;
```

```
const byte ROWS = 4; // Four rows
```

```
const byte COLS = 4; // Four columns
```

```
//Define the keymap
```

```
char keys[ROWS][COLS] = {
```

```
{ '1','2','3','A' },
```

```
{ '4','5','6','B' },
```

```
{ '7','8','9','C' },
```

```
{ '*', '0', '#', 'D' }
```

```
};
```

```
//// Connect keypad ROW0, ROW1, ROW2 and ROW3 to these Arduino pins.
```

```
byte rowPins[ROWS] = {6,7,8,9}; //connect to row pinouts
```

```

// Connect keypad COL0, COL1, COL2 and COL3 to these Arduino pins.
byte colPins[COLS] = {2,3,4,5}; //connect to column pinouts

// Create the Keypad
Keypad keypad = Keypad( makeKeymap(keys), rowPins, colPins, ROWS, COLS );
LiquidCrystal lcd(12, 11, 38,39,40,41);
void setup(){
  lcd.begin(16,2);
  lcd.setCursor(0,0);
  //lcd.print("digital locker");
  lcd.setCursor(0,0);
  lcd.print("Enter Password");
  //delay(5000);
  //lcd.clear();
  Serial.begin(9600);
  pinMode (IN1, OUTPUT);
  pinMode (IN2, OUTPUT);
}

void loop(){
  char key = keypad.getKey();
  if (key != NO_KEY){
    delay(60);
    switch (key){

      default: processNumberKey(key);
    }
  }
}

while (Serial.available())
{
  delay(10);
  char c = Serial.read();
  voice += c;
}
if (voice.length() > 0)
{
  Serial.println(voice);

  if(voice == "on")
  {
    lcd.clear();
    // lcd.print("bluetooth connected");
    digitalWrite(IN1,LOW);
    digitalWrite(IN2,HIGH);
    lcd.setCursor(0,0);
    lcd.print(" Password");
    lcd.setCursor(0,1);
    lcd.print(" Access Granted");
  }
}

```

```

    delay(5000);
    lcd.clear();
    digitalWrite(IN1,LOW);
    digitalWrite(IN2,LOW);
    //lcd.print("access granted");

}

else if(voice == "off")
{
    digitalWrite(IN1,LOW);
    digitalWrite(IN2,LOW);
}

voice="";
}
}

void processNumberKey(char key) {
    Serial.print(key);
    currentPasswordLength++;
    password.append(key);
    if (currentPasswordLength == maxPasswordLength) {
        checkPassword();
    }
}

void checkPassword() {
    if (password.evaluate()){
        digitalWrite(IN1, LOW); // turn the LED on (HIGH is the voltage level)
        digitalWrite(IN2, HIGH);
        lcd.setCursor(0,0);
        lcd.print(" Password   ");
        lcd.setCursor(0,1);
        lcd.print(" Access Granted");
        delay(5000);
        lcd.clear();
        //delay(5000);
        digitalWrite(IN1, LOW); // turn the LED off by making the voltage LOW
        digitalWrite(IN2, LOW);

    } else {

        digitalWrite(IN1, LOW); // turn the LED off by making the voltage LOW
        digitalWrite(IN2, LOW);
        lcd.setCursor(0,0);
        lcd.print(" Password   ");
    }
}

```

```
    lcd.setCursor(0,1);  
    lcd.print(" Access Denied!");  
  
    delay(5000);  
    lcd.clear();  
}  
resetPassword();  
}  
  
void resetPassword() {  
    password.reset();  
    currentPasswordLength = 0;  
}
```